

Branching and boundary tests

Project: Python readiness portfolio

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Boolean conditions

Without opening the old lesson, explain one key idea from Boolean conditions. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Values variables and types

Without opening the old lesson, explain one key idea from Values variables and types. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace the tested path for Branching and boundary tests. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Branching and boundary tests. Proposed explanation: “Every condition in the chain always runs”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Trace branch order

Predict the action for 30, then trace 29 and -1. Explain why validity is checked before the lamp decision.

```
reading = 30

if not 0 <= reading <= 100:
    action = "invalid"

elif reading < 30:
    action = "on"

else:
```

```
    action = "off"

print(action)
```

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Explain the chain

Describe the purpose of if, elif and else in the worked duration sorter. Explain why print(label) is outside the indented branches.

My reasoning and evidence

6. W2 Trace the tested path

For -1, 12 and 13, record checked conditions, skipped conditions and the final label. Run each as a fresh script after predicting.

My reasoning and evidence

7. W3 Repair the overwritten result

Trace the two-independent-if example for -1. Identify which assignment overwrites the earlier label and make the smallest repair.

My reasoning and evidence

8. W4 Build a new sorter independently

Use an integer tiles count. Label a negative value Invalid, zero Empty, 1 through 6 Small and values above 6 Large. Write an ordered if elif else script and predict -1, 0, 1, 6 and 7. Include one print after the chain.

My reasoning and evidence

9. W5 Choose revealing regression tests

A repair changes the small-count condition from < 6 to ≤ 6 . Name the case that exposes the original error and two different kinds of earlier cases to rerun.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled branching and boundary tests evidence sheet to Python readiness portfolio. Show how you can trace the selected branch in an if elif else chain. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Branching and boundary tests

Check invalid values before branching. A toy shelf label uses fewer than five books as "small", otherwise "large", but negative counts are invalid.

Python 3: `if count<0: label="invalid"; elif count<5: label="small"; else: label="large"`. Read as indented branches.

Branching and boundary tests

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Check validity

For count=-2, the first condition selects invalid.



2. Check small boundary

For count=4, validity passes and count<5 selects small.



3. Handle equality

For count=5, both earlier conditions fail, so large is selected.

CHANGE ONE THING

Remove the first negative-count branch.

Explain the changed result using the matching lesson.

Trace -2, 4 and 5 through the branches. Explain why the negative check comes first and what label -2 gets if it is removed.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted outputs and the changed-case explanation. Use actual Python 3 results to check the prediction after reasoning first.

13. Project portfolio milestone

Project: Python readiness portfolio

Current milestone: Python readiness portfolio. Keep runnable code, predictions, actual outputs, one repaired error and an explanation of a function on changed inputs.

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted outputs and the changed-case explanation. Use actual Python 3 results to check the prediction after reasoning first.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Check invalid values before branching. A toy shelf label uses fewer than five books as "small", otherwise "large", but negative counts are invalid.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? For count=5, both earlier conditions fail, so large is selected.

1. Check validity
2. Handle equality
3. Check small boundary

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Handle equality For count=5, both earlier conditions fail, so large is selected. An if/elif/else chain selects the first matching branch. Putting invalid handling first prevents a negative count from being mislabelled as a small valid collection.

Practice 2: explain the deciding evidence

Changed-case practice: Remove the first negative-count branch. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. A count of -2 would incorrectly receive small.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: A count of -2 would incorrectly receive small. Negative two is less than five, but it is not a permitted measured count.

Repair a missing building block

Review Boolean conditions (python-bridge/4). Goal: Translate stated requirements into comparisons. Short practice: Give both comparison results for ages 11, 15 and 16. Which comparison rejects each outside value, and why does and matter?. Return to this week and explain what you changed.

Review Values variables and types (python-bridge/2). Goal: Distinguish integer, floating-point, text and Boolean values. Short practice: Predict both print outputs and explain why they differ. Does count+tag work, and what happens to each valid output after count becomes ten?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

